

Alexander Duan

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EDUCATION

University of California San Diego

GPA: 3.74

Bachelor of Science in Computer Engineering

Aug 2025 – Jun 2029

California Academy of Mathematics and Science

Aug 2021 – June 2025

UW GPA: 3.83, W GPA: 4.44 | *National Merit Commended Scholar*

Carson, CA

Long Beach City College

Sep 2021 – Jun 2025

Java Programming (CS 21), Data Structures and Algorithms (CS 22)

Long Beach, CA

Stanford University

Jun 2024 – Aug 2024

Data Science (DATASCI 112), Electrical Engineering (ENGR 140M)

Palo Alto, CA

EXPERIENCE

Diamond Staff Intern for AI Projects Board

Oct 2025 - Present

Association for Computing Machinery

San Diego, CA

- Developed interactive notebooks using PyTorch and Scikit-learn, to serve as a foundation for future projects, covering end-to-end deep learning workflows from tensor manipulation and data formatting to CNN architecture design and training.
- Represented the AI Projects Board at the ACM Kickoff, effectively pitching technical initiatives to prospective members and driving recruitment for the upcoming project cohort.

FRC Team 687 Robotics — Captain, Programming Lead

Mar 2023 – May 2025

- Led 40+ member team to World Championships (2023, 2024) through strategic planning and cross-departmental coordination
- Developed robot pose estimation algorithms using Java WPILib and Limelight libraries to integrate robot localization
- Improved autonomous repeatability by 65%, raising team Estimated Performance Average (EPA) from top 2000 globally to top 8 in California and top 54 worldwide out of 3400+ teams
- Created comprehensive Summer training schedule and programming curriculum covering necessary tools and applications

PROJECTS

YOLO-Based Fruit Object Detection Model

Oct 2025 – Jan 2026

- Trained custom CNN for fruit detection using YOLO framework and PyTorch, optimizing from COCO's 80+ classes
- Processed COCO2017 dataset with 118K training images, 5K validation images. Extracted and annotated bounding boxes, and engineered a single-box-per-cell architecture, moving from YOLO API to building a custom CNN from scratch
- Built custom loss function, trained predictive model, and validated performance across Google Colab and Kaggle notebooks

LaunchX Entrepreneurship and Innovation

Jul 2023 – Sep 2023

- Founded Travlgo.com: AI-powered platform that creates tailored travel itineraries by leveraging machine learning to analyze user preferences and constraints for personalized trip planning solutions based on user parameters and needs
- Built scalable web architecture, creating API endpoints to connect user inputs to custom ML itinerary generation model.
- Delivered pitch, business plan to investor panel to build support and communicate platform value; drove 300+ site visits

YouTube Programming Channel

Aug 2024 – Present

- Created educational robotics content reaching 70+ subscribers, focusing on robot pose estimation and programming tutorials
- Produced an instructional 8 video series on robotics Limelight camera integration, object detection, and robot localization
 - <https://www.youtube.com/@alexlduan/playlists>
- Hosted a weekend-long computer vision workshop at my school, engaging 80+ participants across 4 robotics teams

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript

Tools/Frameworks: VSCode, Git, UTM, ROS2, Kaggle, Jupyter Notebooks, Google Colab, React, Next.js

Libraries: YOLO (image classification), PyTorch, OpenCV

Areas of Interest: Object detection, Autonomous Systems, PID control algorithms, Sensor/Data integration